Perceptron Loss Function

$$L(W_1, W_2, b) = \frac{1}{n} \sum_{i=1}^{n} 2(y_i, f(x_i)) +$$

$$\alpha R(W_1, W_2)$$

$$\boxed{L = \frac{1}{n} \sum_{i=1}^{n} \max(0, -y_i f(x_i))}$$

n = # of rows in data

Explanation of Loss Function

$$L = \frac{1}{n} \sum_{i=1}^{n} \max(0, -y_i f(x_i))$$

Where

$$f(x_i) = W_1 x_{i1} + W_2 x_{i2} + b$$

Gradient Descent

$$L = \underset{W_1 W_2 b}{\arg\min} \frac{1}{n} \sum_{i=1}^{n} \max(0, -y_i f(x_i))$$

Where

$$f(x_i) = W_1 x_{i1} + W_2 x_{i2} + b$$

$$\frac{\partial L}{\partial W_1} = \frac{\partial L}{\partial f(x_i)} \times \frac{\partial f(x_i)}{\partial W_1}$$

$$\frac{\partial L}{\partial f(x_i)} = \begin{cases} 0 & \text{if } y_i f(x_i) \geq 0 \\ -y & \text{if } y_i f(x_i) < 0 \end{cases} \quad \frac{\partial f(x_i)}{\partial W_1} = x_{i1}$$

$$\frac{\partial L}{\partial W_1} = \begin{cases} 0 & \text{if } y_i f(x_i) \geq 0 \\ -y_i x_{i1} & \text{if } y_i f(x_i) < 0 \end{cases}$$

$$\partial L = \begin{cases} 0 & \text{if } y_i f(x_i) \geq 0 \end{cases}$$

$$\frac{\partial L}{\partial W_2} \quad \begin{cases} -y_i x_{i2} & \text{if } y_i f(x_i) < 0 \end{cases}$$

$$\frac{\partial L}{\partial b} = \begin{cases} 0 & \text{if } y_i f(x_i) \geq 0 \\ -y_i & \text{if } y_i f(x_i) < 0 \end{cases}$$